

CSE340: Computer Architecture

Assignment 3

Chapter 3

Note: For the questions below, if necessary, Consider 5 digits while performing the decimal to binary conversion of the floating portion.

Question 01

Convert the following IEEE-754 number into a decimal number where 6 bits are allocated for the exponent field in the representation.

0x ABB9609

Question 02

Perform the arithmetic operations using the Floating point format

- a. $50.7869 + 79.83 - 29.58$
- b. $64.2486 * 49.1832$

Question 03

Subtract -4.0210 from 28.4810 using IEEE-754 single-precision floating-point representation. Check if the result has overflow or underflow or none.

Note: Consider 5 digits while performing the decimal to binary conversion of the floating portion.

Question 04

- a. Why is a bias added to the actual exponent in the IEEE 754 floating-point representation, and how does this affect the encoding of both positive and negative exponents?
- b. How does optimized multiplication improve efficiency and performance compared to traditional long multiplication, especially in terms of speed and computational complexity?

Question 05

	Instruction name
1	mul
2	mulh
3	mulhu
4	mulhsu

Explore these instructions from the slide first.

You **must** answer the following questions for the explanation of each of the instructions mentioned above:

- (i) What do they do/ How do they work?
- (ii) Explain their syntax.
- (iii) Write an example for each of the instructions.

Question 06

fld x5, f3, f5

- (i) Why x5 is an integer register?
- (ii) f3 and f5 both are single precision numbers. True or False. If false, write the correct statement.
- (iii) after running the instruction you see that x5 register contains 0. What can you comment on?

Question 07

Write the necessary RISC-V code to compare two floating point registers f1 and f2. If f1 and f2 are equal jump to a label called “**jumpEqual**” else jump to another label called “**jumpNotEqual**”

Note: both the numbers stored in f1 and f2 registers are double-precision floating point numbers.

Question 08

Write the necessary RISC-V code to compare two floating point registers f1 and f2. If f1 is greater than f2, jump to a label called “**jumpGreater**” else jump to another label called “**jumpNotGreater**” and divide the variable c with 16 and store that answer in c.

Note: both the numbers stored in f1 and f2 registers are double-precision floating point numbers and variable c is stored in x19.